

Preparation of the sandbox

1. Use the same host (nymfe) as last time.
2. Log in to the host and go to your local copy of the `labs` directory.
3. **Update the repo** by running `git pull` in `labs` directory. Go to `2` directory.
4. Login using your **xlogin** and **faculty password** to get access to Podman images:

```
podman login gitlab.fi.muni.cz:5050
```
5. Run `podman-compose pull` to **download images** used in this lab.
6. Log into the **INJECT Exercise Platform** at <https://inject3.fi.muni.cz/>.
7. Enter **Lab 2** exercise in the platform.
8. Go to Sandbox configuration and download the `config.env` file.
9. Move the file to `labs/2` directory and rename it to `.env`.

PB177 Cyber Attacks

Seminar 2 – Initial access, execution

Autumn 2026

Last seminar...

- We covered **network enumeration** and how to **gather information** about targets.
- We practiced using **nmap** for network scanning and **dirb** for subdirectory discovery.
- We also introduced **generative AI** and its role in enhancing social engineering and phishing attacks.
- All of these tools and techniques prepare us for the **next step: exploiting discovered vulnerabilities**.

Learning objectives

At the end of this lab session, you will be able to:

- use **Metasploit** for **initial access** and **execution** using one particular vulnerability
- use **Wireshark** for **analyzing packets** generated by Metasploit
- **detect** the particular exploit and **mitigate** the vulnerability

Seminar agenda

1. **KvIS**
2. “*Breaking & Entering 101*” – Initial Access & Execution
3. “*Your New Best Friend (or Worst Enemy?)*” – Metasploit
4. “*See the Traffic, See the Truth*” – Wireshark
5. “*Hack It Till You Make It*” – hands-on exercises on your machines

KvIS

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2. ***“Breaking & Entering 101”*** – Initial Access & Execution
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What is Initial Access?

- **First step** attackers take to get inside a system
- Often achieved through **phishing, exploiting vulnerabilities, or credential theft**
- Without initial access, attackers can't proceed to further stages
- Examples:
 - Clicking a malicious link
 - Exploiting a public-facing service
 - Using leaked passwords

What is Execution?

- **Running malicious code** after gaining initial access.
- This is where the attacker starts doing real damage or moving deeper into the system
- Examples:
 - Launching a payload via Metasploit
 - Running a malicious script
 - Triggering code through a vulnerability

What is the difference?

- **Initial Access** = how attackers get in
- **Execution** = what attackers do after getting in
- In simple terms (house analogy):
 - Last time, we learned how to **find attack vectors** – like doors, windows, or chimneys (= Reconnaissance)
 - **Initial access** is when you ***unlock the door or open the window from the outside***
 - **Execution** is the following stage where you ***enter the house and act – like stealing valuables***



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Metasploit Framework

- What is it?
 - A powerful **exploitation framework** used for **developing, testing, and executing exploits** against target systems
- What can it do?
 - **Discover vulnerabilities**
 - **Execute exploits to gain access**
 - Generate payloads (such as reverse shells)
 - Manage sessions with compromised machines
 - Perform post-exploitation tasks
- Why do we use it?
 - Speeds up the exploitation process
 - Provides a structured and modular way to test security
 - Widely used in ethical hacking, penetration testing, and security training

Keywords:

- vulnerability
- exploit
- payload
- reverse shell
- session



Tool usage

- See [cheat sheet](#) in Study materials in IS.

Seminar agenda

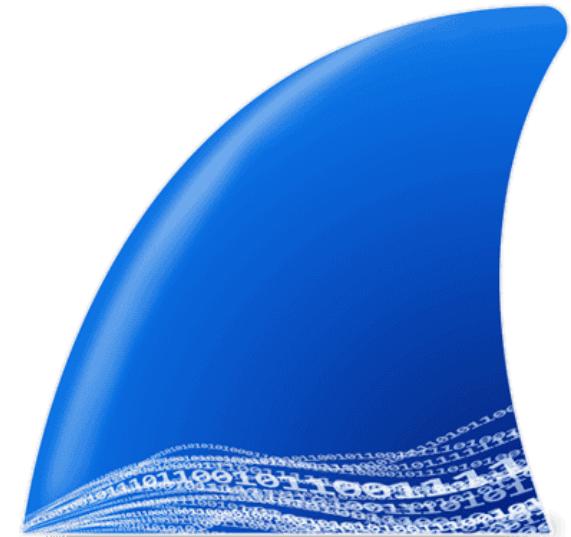
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Wireshark

- What is it?
 - A **network protocol analyzer** that lets you capture and inspect network traffic in real time
- What can it do?
 - Capture live network packets
 - Analyze communication between devices
 - Filter and search for specific protocols or IPs
 - Help identify suspicious or malicious traffic
 - Reconstruct network sessions (such as file transfers or logins)
- Why do we use it?
 - To **understand what's happening on the network**
 - To **troubleshoot network issues**
 - To **detect attacks or data leaks**
 - To **visualize traffic** in a way that's easy to follow and learn from

Keywords:

- network protocol
- capture
- communication
- traffic



Usage of Wireshark

- Being proficient with Wireshark takes time and practice
- You should already be familiar with the basics from **PB156cv** and **PV080**
- In this seminar, you will mainly use Wireshark to:
 - **Capture traffic from your own machine**
 - **Filter packets** to find useful information
 - **Identify payload delivery and initial access steps** in a simulated scenario

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Now that you know your toolkit...

- It's time to put **theory** into **practice**!
- In the following part of the seminar, you'll **try out the tools** you've just learned about.
- You'll explore **basic usage**, **test real-world scenarios**, and **get a feel** for how these tools **behave in action**.
- Don't worry if you make mistakes, this is where learning really begins.

Start the lab in INJECT

- The platform will display tasks that you will complete in your sandbox or using a browser.
- Work at your own pace.
- Feel free to **use “Get help”** for each task or **show the solution** on the platform, ask CyberBrain or the tutor.
- If you do not complete the lab in class, complete it at home in 2 weeks.
- After that, we will check whether you finished and award points for completion.

Start the sandbox

Provided you have completed all steps at the very first slide, continue with these steps:

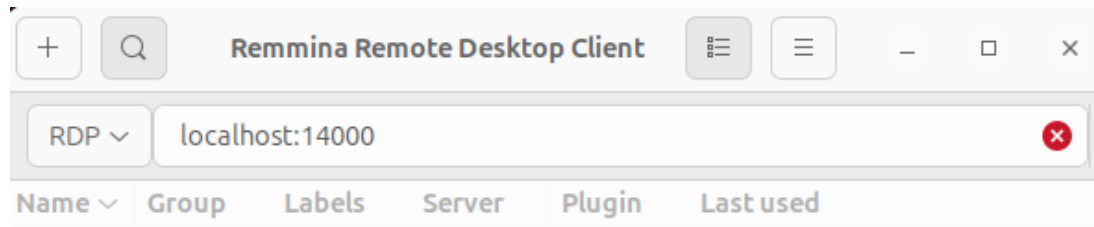
1. Run `podman-compose up -d` to **start containers**.
2. Check that **3** containers are running: `podman-compose ps`
3. Run `podman-compose exec attacker bash` to get root **shell** of **attacker** machine:

```
xvykopa1@nymfe105:~/pb177/labs/1$ podman-compose exec attacker bash
root@attacker:~#
```

4. Connect to the GUI of the **attacker** machine (see next slide).

Access to attacker using GUI

- Connect to the graphical interface of the **attacker** using the **Remmina** remote desktop client
 - Open **Remmina** – either using an icon or run `remmina` in the terminal
 - Connect as user `kali` via remote desktop running at `localhost` at port `14000`



- Alternatively, there is a script prepared for you in the **labs** repository in the first lab directory
 - In your terminal, run `open ../1/attacker-rdp.remmina`
- The password is `kali` (the same as the username)

Finishing labs at nymfe from home

- You can run the sandbox at “your” nymfe at A219 or in PC hall and connect to it from home using VPN and SSH port forwarding.
- See the [corresponding section in our cheat sheet](#).
- Please use “your” nymfe in A219 only when no seminars are running there, see the schedule of the room in IS MU.
- Alternatively, use nymfe001–010 in the PC hall.

Note where you are seating

- Please use the same nymfe next time to save disk space
- Make a note or take a photo of the nymfe number, such as
nymfe89

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